

Project Design Phase

Solution Architecture

Date	3 July 2026
Team ID	
Project Name	Rising Waters
Maximum Marks	4 Marks

Solution Architecture:

The solution architecture defines how the proposed system uses Machine Learning and web technologies to provide accurate flood risk prediction. It connects user requirements with a practical and scalable technical solution.

The main goals of the solution architecture are to:

- **Develop an intelligent flood prediction system** using historical weather and rainfall data.
- **Provide a clear architecture** showing the interaction between the user interface, Flask application, Machine Learning model, and data sources.
- **Define the application's features**, including data input, preprocessing, flood prediction, and result generation.
- **Support early warning and disaster preparedness** by providing accurate and timely flood predictions.
- **Provide a scalable architecture** that can be extended with cloud deployment, real-time weather APIs, IoT sensors, GIS mapping, and notification services.
- **Ensure the application is reliable, easy to maintain, and user-friendly** for disaster management authorities and the general public.
- **Provide implementation guidelines** for developing, testing, deploying, and maintaining the Flood Prediction System.

Example - Solution Architecture Diagram:

